

Bayesian gamma processes for optimizing condition-based maintenance under uncertainty

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The aim of this article is twofold: (i) modeling partially observed crack growth of industrial components using gamma processes and (ii) providing estimators of the best maintenance time in a statistical Bayesian framework. The choice of a Bayesian framework is motivated by the small size of data, the availability of expert knowledge about the crack propagation, and more generally, the concern about the integration of parametrical uncertainties when optimizing a maintenance action. The article answers to the methodological question of Bayesian prior elicitation by adopting a strategy based on virtual data information and defines optimal replacement times as posterior Bayes estimators minimizing appropriate cost functions. More precisely, the industrial data are described, and two different levels of available information are considered. Then, the Bayesian parameter estimation procedure in each case is thoroughly explained, by conducting MCMC runs. Different criteria for maintenance optimization, taking account all uncertainties, are considered and discussed. The overall procedure is tested on simulated data and applied over a real dataset. Copyright © 2014 John Wiley & Sons, Ltd.

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1. Introduction

One of the challenges in maintenance optimisation is to propose the most effective balance between preventive actions, which avoid the occurrence of failures but reduce the lifecycle duration of the maintained component, and corrective actions carried out after failure. Each maintenance action incurs a cost and very often the preventive maintenance conducted on an operational system is cheaper than a corrective maintenance action. Above all, a substantial cost is involved during the unavailability period, the duration between the failure time and the corrective action. The main aim is, henceforth, to avoid costly maintenance actions and also to reduce the period of unavailability while maximizing the system lifetime.

For such a task, it is needed to predict accurately the failure of the system. In this purpose, the degradation process or the lifetime may be modeled and statistically assessed using prior knowledge and legacy (e.g., historical) data collected on similar systems. Then the fitted model can be used for prediction, while taking account of estimation uncertainties. Failure times can only be used to estimate lifetime models, regardless to its deterioration level, cf. [1–3]. In more informational situations, a deterioration indicator [4–6], which measures dynamically through time the degradation level, may be proposed. Such a model should, on the one hand, take account of the different properties of the deterioration process and, on the other hand, be tractable for failure prediction.

Because in similar running conditions two identical systems can undergo different deteriorations, the degradation process is often considered as a random phenomenon. In such cases, in an aim of modeling and prediction, the deterioration is considered as a stochastic process [7–10]. When the system is submitted to aging and the deterioration level is a monotone, positive and continuous function of time, summarized by a scalar variable, several kinds of stochastic processes appear

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appropriate for this task. Classical degradation models belong to the family of Lévy processes. Among them, the two mainly used models is the Wiener process (with positive drift) and the gamma process. The Wiener process is especially adapted to cases where measurement errors can induce apparent local decreasings of the deterioration level [11], while the gamma process is sometimes rather preferred because it has increasing sample paths. Indeed, the gamma process is often used to describe the gradual deterioration caused by accumulation of wear [12–16]. It can be efficiently used to describe continuous natural phenomena as erosion, corrosion, and concrete creep [17–20]. Another interest of the gamma process is the existence of an explicit probability distribution function, which permits feasible mathematical developments. In the case that motivated this study, the level variable under consideration is the size of a crack appearing on an industrial component; for this very reason, Blain *et al.* [21] proposed a gamma process for the crack size modeling.

A first issue encountered in practice is the assessment of its parameters. Frequentist statistical methods can be used to estimate them if there is a substantial number of observations [8, 22]. However, small crack sizes are often unseen by measurement means (for instance ultrasonic means) because they are overwhelmed by observational noise. Again, the motivating case study considered further belongs to this class of problems. In such cases, choosing a Bayesian framework is relevant to improve the robustness of the estimation provided coherent expert knowledge is available [23–25]. The ability of Bayesian approaches to manage uncertainties was highlighted in [26] in the context of reliability assessment using degradation models. Conducting an informative Bayesian study implies a first step of *prior elicitation*, namely defining a prior distribution that encodes properly the expert information. This task is difficult since submitted to subjective choices. Therefore, it is essential that the main features of the elicited prior be understandable to experts, reliability engineers, and decision makers, as pointed out in [27]. Bayesian frameworks were explored for homogeneous or non-homogeneous Wiener processes with positive drifts in [28–30]. A major feature of modeling, these authors consider the drifts as random with Gaussian priors and produce adequate estimation methods. As far as gamma processes are concerned, the use of an objective Bayesian framework was explored in [31] only for determining the maximum likelihood estimator of a bivariate process in an indirect way. But to our knowledge, there is today no methodology of informative prior elicitation adapted to gamma processes.

A second issue is establishing clear and simple rules for estimating the best maintenance time. Often, the system is assumed to be in a failure state when the aging variable is greater than a predetermined threshold even if it is still functioning, which implies a corrective maintenance. In this situation, its high level of deterioration is unacceptable either for economic reasons (poor products quality, high consumption of raw material, etc.) or for safety reasons (high risk of hazardous breakdowns or service disruption). In such a perspective, van Noortwijk and van Gelder [13] considered for instance the problem of maintaining coastal flood barriers. A preventive maintenance operated before reaching this failure state can be less costly, provided the time chosen for it realizes a trade-off between the two costs and the number of systems to visit.

The decision-theoretic foundations of the Bayesian framework allow to produce estimators of optimal maintenance times with desirable order properties [23–25], by minimizing cost functions averaged over all posterior odds. Optimality rules have to be defined to take account of the costs of being too pessimistic (replacing too early) or optimistic (replacing too late), or the possibility of replacing two components in the same time with a lower cost than replacing each at different times. The formulation of such rules would normally involve many actors (decision makers), but in practice, the reliability engineer or the statistician has to provide pointwise estimates with little information available [32, 33].

The present article aims at providing usable solutions to both issues. First, a specific methodology of prior elicitation is considered, which makes the prior features understandable for the reliability engineer and the decision maker who uses the posterior results. According to this formulation, the expert is perceived as a provider of virtual data. This elicited prior appears especially suitable for sensitivity analyses under the constraint that the posterior result remains mainly driven by the (objective) data information. In a second step, once the Bayesian estimation is performed by conditioning the prior distribution to the observation of available data using specific numerical methods (Monte Carlo Markov Chains), several simple and classical choices of cost functions are considered to complete the analysis and provide optimal maintenance times in usual situations.

Therefore, the remainder of the paper is organized as follows. Section 2 gives a description of the considered data, which are assumed to suffer from strong right and left censoring, and their link with the gamma process. The Bayesian modeling, focused on the prior elicitation task and the posterior numerical computation, is detailed in Section 3, which besides make use of simulated experiments. Then, Section 4 deals with the practical use and choice of various cost functions to define appropriate maintenance decision rules. The complete methodology is finally applied on the real (motivating) case study in Section 5, while a discussion section ends this article by summarizing the main results and opening avenues for future research.

2. Data and deterioration model

Inspired by the case study detailed in Section 5, let us consider a set of M identical components submitted to intensive effort. It is assumed that the deterioration of a individual component $k \in \{1, \dots, M\}$ is characterized by a crack size $X_{k,t}$, observed at inspection times $t \in \{t_{1,k}, \dots, t_{n_k,k}\}$ (to alleviate notations, we remove the second index k for the inspection times). $X_{k,t}$ is therefore positive and monotonously increasing through time. When they appear, cracks are not observed until reaching a size z because of the limitations of observational (e.g., ultrasonic) processes. The failure is assumed to occur from a size $\mu > z$. For the reasons evoked earlier, $X_{k,t}$ is assumed to be a stochastic gamma process with following features:

- for all k , $X_{k,0} = 0$, and $(X_{k,t})_{t \geq 0}$ has independent increments $Z_{k,i} = X_{k,t_i} - X_{k,t_{i-1}}$;
- for all k and $0 \leq s \leq t$, the increment $X_{k,t} - X_{k,s}$ follows a gamma probability distribution function with density function

$$f_{\alpha(t-s), \beta}(x) = \frac{1}{\Gamma(\alpha(t-s))} \cdot \frac{x^{\alpha(t-s)-1} e^{-\frac{x}{\beta}}}{\beta^{\alpha(t-s)}} \mathbb{1}_{\{x \geq 0\}}. \quad (1)$$

Let us recall that the average deterioration rate between s and t , for $0 \leq s \leq t$, is $\alpha \cdot \beta(t-s)$, and the variance of the deterioration rate is $\alpha \cdot \beta^2(t-s)$.

Denote now t_{r_k} the last inspection time before the first determination of the true crack size, that is,

$$t_{r_k+1} = \min_{i \in \{1, \dots, n_k\}} \{t_i, X_{k,t_i} \geq z\}.$$

The data likelihood incorporates two censoring terms function of t_{r_k} .

- (1) **Left-censoring term.** The likelihood term due to the non-observation of the crack size between $t_0 = 0$ and t_{r_k} for the component k is the following probability (assuming $X_{k,t_0} = 0, \forall k \in \{1, \dots, M\}$)

$$P(X_{k,t_1} \leq X_{k,t_2} \leq \dots \leq X_{k,t_{r_k}} < z) = P\left(\sum_{i=1}^{r_k} Z_{k,i} < z\right)$$

with

$$\sum_{i=1}^{r_k} Z_{k,i} \sim \mathcal{G}(\alpha t_{r_k}, 1/\beta) \quad (2)$$

where $\mathcal{G}(a, b)$ denotes the gamma distribution with shape parameter a and mean a/b . Therefore, one has

$$P(X_{k,t_1} \leq X_{k,t_2} \leq \dots \leq X_{k,t_{r_k}} < z) = \frac{\gamma(\alpha t_{r_k}, \frac{z}{\beta})}{\Gamma(\alpha t_{r_k})}$$

where $\gamma(a, b) = \int_0^b x^{a-1} e^{-x} dx$ is the lower incomplete gamma function.

- (2) **Right-censoring term.** The likelihood term due to the observation of $\tilde{Z}_{k,r_k+1} = X_{k,t_{r_k+1}} - z$, which is a lower bound for the true Z_{k,r_k+1} , is

$$P(Z_{k,r_k+1} > x_{k,t_{r_k+1}} - z) = \frac{\Gamma(\alpha(t_{r_k+1} - t_{r_k}), (x_{k,t_{r_k+1}} - z)/\beta)}{\Gamma(\alpha(t_{r_k+1} - t_{r_k}))}$$

where $\Gamma(a, b) = \int_b^\infty x^{a-1} e^{-x} dx$ is the upper incomplete gamma function for $x \geq 0$ and $a > 0$.

Finally, the likelihood of data $\mathbf{d}_k$ for the k th component is

- if there is no exact measure,

$$\ell_k(\alpha, \beta | \mathbf{d}_k) = \frac{\gamma(\alpha t_{n_k}, \frac{z}{\beta})}{\Gamma(\alpha t_{n_k})};$$

- if there is only one exact measure,

$$\ell_k(\alpha, \beta | \mathbf{d}_k) = \frac{\gamma\left(\alpha t_{n_k-1}, \frac{z}{\beta}\right) \Gamma\left(\alpha(t_{n_k} - t_{n_k-1}), (x_{k,t_{n_k}} - z)/\beta\right)}{\Gamma(\alpha t_{n_k-1}) \Gamma(\alpha(t_{n_k} - t_{n_k-1}))};$$

- if there is more than one exact measure,

$$\begin{aligned} \ell_k(\alpha, \beta | \mathbf{d}_k) &= \frac{\gamma\left(\alpha t_{r_k}, \frac{z}{\beta}\right) \Gamma\left(\alpha(t_{r_k+1} - t_{r_k}), (x_{k,t_{r_k+1}} - z)/\beta\right)}{\Gamma(\alpha t_{r_k}) \Gamma(\alpha(t_{r_k+1} - t_{r_k})) \prod_{i=r_k+2}^{n_k} \Gamma(\alpha(t_i - t_{i-1}))} \\ &\times \beta^{-\alpha(t_{n_k} - t_{r_k+1})} \exp\left\{-\frac{1}{\beta} \sum_{i=r_k+2}^{n_k} z_{k,i}\right\} \left[\prod_{i=r_k+2}^{n_k} z_{k,i}^{t_i - t_{i-1}}\right]^\alpha. \end{aligned}$$

Then the likelihood $\ell(\alpha, \beta | \mathbf{d})$ of all observed data $\mathbf{d}$ is the product of the $\ell_k(\alpha, \beta | \mathbf{d}_k)$ over the $k = 1, \dots, M$ identical components.

3. Bayesian estimation framework

A Bayesian framework is chosen here to carry out the estimation of $\theta = (\alpha, \beta) \in \mathbb{R}_+^2$ and any function of interest $h(\theta)$ (e.g., the best maintenance time defined hereafter). Namely the parameter vector θ is considered as random, endowed with a prior distribution of density $\pi(\theta)$, and the estimation task consists in obtaining a detailed description of its posterior distribution $\pi(\theta | \mathbf{d}) \propto \pi(\theta) \ell(\theta | \mathbf{d})$, having learned from the data $\mathbf{d}$ through $\ell(\theta | \mathbf{d})$.

3.1. Objective Bayesian framework

3.1.1. Modeling. To avoid the well-known issues related to the difficult elicitation of a prior distribution [34, 35], an objective Bayesian framework can be chosen [36–39]. Especially defended by Clarke [40], a relevant non-informative prior measure $\pi(\theta)$ is Jeffreys' prior. It is basically defined as the square root of the determinant of the Fisher matrix and is therefore intrinsically linked to the form of the likelihood (see [36] for details). To simplify the elicitation and because the posterior distribution is driven by the data rather than the prior in any objective Bayesian framework, the censoring aspects of the likelihood are not taken into account (but discussed further in Section 6). To be independent on the scale effect imposed by the heterogeneity of time steps along a cracking trajectory and between the components, the prior is defined for a standard gamma distribution, that is, for the likelihood of the first occurrence of the process Z . From [41], one has

$$\pi(\alpha, \beta) \propto \frac{1}{\beta} \sqrt{\alpha \Psi_1(\alpha) - 1} \quad (3)$$

where $\Psi_1(\cdot)$ is the trigamma function. It must be noticed that the posterior obtained from this prior and the first likelihood element plays the role of the prior for the second likelihood element, and so on. Because $\alpha \Psi_1(\alpha) - 1 > 0$ for $\alpha > 0$, the prior is well defined.

3.1.2. Posterior computation. Sampling from the posterior distribution, by using specific numerical techniques as Monte Carlo Markov Chains (MCMC), is needed because the posterior distribution is not of tractable form. However, it is possible to take into account the missing data structure of the problem to implement a Gibbs algorithm [25, 42] taking benefit from conditional conjugation properties. Indeed, if all missing statistics $S_k = \sum_{i=1}^{r_k} Z_{k,i}$ could be known (available or reconstituted using (2) under the constraint $S_k \leq z$), the posterior distribution of β conditional to α and observed data $\mathbf{d}$ would be inverse gamma (denoted by IG):

$$\beta | \alpha, \mathbf{d} \sim IG\left(\alpha \sum_{k=1}^M t_{n_k,k}, \sum_{k=1}^M \left[S_k + \sum_{i=r_k+1}^{n_k} z_{k,i}\right]\right).$$

Because no such feature can be exhibited for the conditional posterior distribution of α , a Metropolis-Hastings mechanism can be inserted within each Gibbs step to produce an ergodic Markov chain [25], using (for instance) a random walk defined with a parameter ρ calibrated from testings. The resulting hybrid algorithm is described hereinafter. Numerical tests conducted to choose a permanent value of $\rho = 0.25$ to ensure a good exploration of the parameter space.

Posterior simulation algorithm 1 (*step* $j - 1 \rightarrow j$).

- (1) Sample $\bar{S}_j \sim \mathcal{G}\left(\alpha_{j-1} \sum_{k=1}^M t_{r_k}, 1/\beta_{j-1}\right) \cdot \mathbb{I}_{\{0 \leq S \leq M \cdot z\}}$.
- (2) Sample $\beta_j \sim IG\left(\alpha_{j-1} \sum_{k=1}^M t_{n_k}, \bar{S}_j + \sum_{k=1}^M \sum_{i=r_k+1}^{n_k} z_{k,i}\right)$.
- (3) Sample $\tilde{\alpha}_j \sim \rho(\cdot | \alpha_{j-1}) \equiv \mathcal{N}\left(\alpha_{j-1}, \rho \cdot \alpha_{j-1}^2\right)$.
- (4) Sample $u_j \sim \mathcal{U}[0, 1]$ and accept $\alpha_j = \tilde{\alpha}_j$ if $u_j \leq \eta_j$ where

$$\eta_j = \min \left\{ 1 \wedge \frac{\ell(\beta_j, \tilde{\alpha}_j | \mathbf{d}) \pi(\beta_j, \tilde{\alpha}_j) \rho(\alpha_{j-1} | \tilde{\alpha}_j)}{\ell(\beta_j, \alpha_{j-1} | \mathbf{d}) \pi(\beta_j, \alpha_{j-1}) \rho(\tilde{\alpha}_j | \alpha_{j-1})} \right\}$$

else accept $\alpha_j = \alpha_{j-1}$.

In the numerical applications considered hereafter, the burn-in periods and stationarity of chains were managed using the classical diagnostics provided in [43], implemented within the CODA software [44]. The MCMC chains were decorrelated until reaching a maximal autocorrelation of 5% between the elements of the chains.

3.1.3. First numerical experiments. In this objective framework, first numerical experiments were conducted using simulations for various representative values of (M, n) and some censoring percentages (implied by arbitrary values of z). The details of the estimation results are presented in Tables A.1–A.4 within Appendix A. The values express the relative bias and relative mean square error, both in percentage, of the usual posterior estimates of (α, β) (posterior means).

In situations close to those encountered in industrial studies, namely when the number of observed components is larger than the size of observed samples per component, the sensitivity of these classical Bayesian estimators to the censoring percentage can lead to tricky inferences about the future failure times. This especially appears when some decision rule (Section 4) is chosen such that the knowledge of the full posterior distribution, namely the mixture of epistemic and aleatory uncertainties on parameters [45], is taken into account to establish what could appear as a conservative management choice for replacing a critical component. For this reason, the knowledge of industrial expert opinions can yield a valuable contribution to the estimation of parameters and functions of interest, in addition to the data information. This contribution to the inference, through an informative Bayesian framework, is explored in the next paragraph.

3.2. Subjective (informative) Bayesian framework

As evoked earlier, the elicitation of an informative prior density $\pi(\theta)$ is rather difficult because it usually introduces a level of subjectivity within the model. Nonetheless, the availability of supplementary information, typically provided by industrial experts, can sometimes be as much valuable as the observations, which are scarce due to the limits of the detection process and the rarity of visits. For this reason, a clear methodology of prior elicitation must be provided, under the form of a (hyper)parametric model for $\pi(\theta)$, the (hyper)parameters of which having practical meanings for the practitioners (experts and reliability engineers). To our knowledge, no work was published about the elicitation of parametric prior distributions on parameters of gamma processes, in the field of industrial reliability. Most researchers (working in this field or not) considered more willingly gamma distribution themselves as prior structures for nonparametric or semi-parametric intensity models [46–48], or used a nonparametric framework to propose Bayesian estimators of gamma process parameters [49]. In the following, we provide a rationale for eliciting a suitable prior form in reliability experiments.

Following an idea originally expressed by Lindley [50], we suggest to consider that a perfect expert opinion should be, roughly speaking, similar to a real data survey and provide an independent and identically distributed (i.i.d.) *virtual sample* of crack increases $\tilde{\mathbf{z}}_m = (\tilde{z}_1, \dots, \tilde{z}_m)$ observed at times $\tilde{\mathbf{t}}_m = (\tilde{t}_1, \dots, \tilde{t}_m)$. Renaming $\pi^J(\theta)$ the non-informative prior elicited earlier, well recognized as a formal representation of ignorance, the ideal informative prior should be the posterior

$$\pi(\theta) = \pi^J(\theta | \tilde{\mathbf{z}}_m, \tilde{\mathbf{t}}_m). \quad (4)$$

Priors build as virtual data posteriors present some advantages in subjective Bayesian analysis. First, they are unique because they are defined only by π^J and the likelihood. Second, the correlation between components of θ is automatically assessed through the Bayes rule, coherently with the sampling model. Third, in conjugate cases, the meaning of prior hyperparameters is clear: they can be expressed as sizes and mean of virtual data samples that summarizes the “strength” and the quantitative value of prior information, which can be directly compared with those characterizing real data (see for instance [51], ... 5.3.3, and [34]). Finally, the aggregation of independent expert opinions is simply carried out through successive Bayes rules, the consensus virtual sample being an aggregation of all virtual data. This avoids choosing opinion pooling rules that can suffer from paradoxes [52]. More details about such elicited priors can be found in [37, 40, 53] and [54].

In the present case, no full conjugacy appears. However, we propose an approximate solution by defining $\pi(\theta)$ as a mixture of inverse gamma ($\mathcal{IG}$) and gamma ($\mathcal{G}$) distributions, linked by the following conditioning:

$$\beta|\alpha \sim \mathcal{IG}(\alpha m \tilde{t}_{e,1}, m \tilde{z}_e), \quad (5)$$

$$\alpha \sim \mathcal{G}(m/2, m \tilde{t}_{e,2}) \quad (6)$$

where the hyperparameters $(m, \tilde{t}_{e,1}, \tilde{t}_{e,2}, \tilde{z}_e)$ must be estimated in function of the available expert information, suggested by the subscript e . To connect the idealistic prior elicitation (4) to (5) and (6) and provide useful meanings for the hyperparameters (by homogeneity with observable quantities Z and T), denote

$$\begin{aligned} \tilde{t}_{e,1} &= \frac{1}{m} \sum_{i=1}^m \tilde{t}_i, \\ \tilde{z}_e &= \frac{1}{m} \sum_{i=1}^m \tilde{z}_i, \\ \tilde{t}_{e,2} &= \frac{1}{m} \sum_{i=1}^m \tilde{t}_i \log \frac{\sum_{j=1}^m \tilde{z}_j / \tilde{z}_i}{\sum_{j=1}^m \tilde{t}_j / \tilde{t}_i}. \end{aligned}$$

Then, given the benchmark prior (3), the posterior (4) is conditionally defined by (5) and

$$\pi(\alpha | \tilde{z}_m, \tilde{\mathbf{t}}_m) \propto \exp \left(-\alpha \left\{ \sum_{i=1}^m \tilde{t}_i \log \frac{\sum_{j=1}^m \tilde{z}_j}{\tilde{z}_i} \right\} \right) \frac{\Gamma(m \alpha \tilde{t}_{e,1})}{\prod_{i=1}^m \Gamma(\alpha \tilde{t}_i)} \sqrt{\alpha \Psi_1(\alpha) - 1}. \quad (7)$$

Using exact Stirling’s formula [55],

$$\Gamma(x) = \sqrt{2\pi} x^{x-1/2} \exp(-x + v(x))$$

where $v(x) = \gamma/(12x)$ and $\gamma \in [0, 1]$, one has

$$\frac{\Gamma(m \alpha \tilde{t}_{e,1})}{\prod_{i=1}^m \Gamma(\alpha \tilde{t}_i)} \propto \alpha^{\frac{m-1}{2}} \exp \left(\alpha \sum_{i=1}^m \tilde{t}_i \log \frac{\sum_{j=1}^m \tilde{t}_j}{\tilde{t}_i} \right) \exp \left(-\frac{\gamma}{12\alpha} \left\{ \sum_{i=1}^m \frac{1}{\tilde{t}_i} - 1/\tilde{t}_{e,1} \right\} \right)$$

and the last term of the product remains close to 1 if the time scale is chosen over the unity. Furthermore, an asymptotic Laurent development gives

$$\sqrt{\alpha\Psi_1(\alpha)-1} = \frac{1}{\sqrt{2\alpha}} \left(1 + \sum_{k=1}^{\infty} \frac{B_{2k}}{\alpha^{2k-1}} \right)$$

where the B_{2k} are the Bernoulli-Faulhaber numbers of the second kind [56]. Therefore, a first-order approximation of the posterior distribution of density (7) is given by the choice (6), provided $\tilde{t}_{e,2} > 0$.

The prior form (5) and (6) is convenient because the meaning of hyperparameters can help the practitioner to provide understandable ranges for conducting posterior sensitivity and robustness studies, as advocated (among others) by Rios Insua *et al.* [27]; $\tilde{t}_{e,1}$ and $\tilde{z}_e$ have the meaning of average time of observation and average crack increase, defining a virtual sample of size m , while $\tilde{t}_{e,2}$ can be (less easily) connected to the speed of cracking. Another interesting feature, which will be useful for treating the case study considered in Section 5, is given in next proposition (proven in the Appendix). The most probable prior value for the increasing of a crack, during any period of time, is independent of $\tilde{t}_{e,2}$. This highlights that $\tilde{t}_{e,2}$ can be seen as a ‘tuning’ parameter, helping to get a flexible prior in the limits allowed by assuming that the prior information remains coherent with the sampling model.

Proposition 1

Denote $r_{\alpha,\beta}(\Delta_i) = \alpha\beta(t_{i+\Delta_i} - t_i)$ the mean crack increasing during the time interval Δ_i . This random variable admits a unique prior modal value (namely, a most likely a priori value)

$$\hat{r}(\Delta_i) = \frac{\tilde{z}_e \Delta_i}{\tilde{t}_{e,1}}. \quad (8)$$

Finally, this prior remains conditionally conjugate for β , which simplifies the final inference given the real observations. Indeed, given the available complete observed sample $\mathbf{d}$ considered earlier mixing right-censored, left-censored, and correct data, the posterior distribution is hierarchically defined by

$$\beta|\alpha, \mathbf{d} \sim \text{IG} \left(\alpha \left\{ m\tilde{t}_{e,1} + \sum_{k=1}^M t_{n_k,k} \right\}, \left\{ m\tilde{z}_e + \sum_{k=1}^M \left[S_k + \sum_{i=r_k+1}^{n_k} z_{k,i} \right] \right\} \right), \quad (9)$$

and

$$\pi(\alpha|\mathbf{d}) \propto \int_0^\infty \{ \ell(\beta, \alpha|\mathbf{d}) \pi(\beta|\alpha, \mathbf{d}) d\beta \} \pi(\alpha)$$

which is not of closed form. Therefore, the posterior simulation algorithm previously presented must be adapted. Again, a value of $\rho = 0.25$ was found practical to ensure good performances when treating the case studies considered later in this article.

Posterior simulation algorithm 2 (step $j - 1 \rightarrow j$).

- (1) Sample $\tilde{S}_j \sim \mathcal{G} \left(\alpha_{j-1} \sum_{k=1}^M t_{r_k}, 1/\beta_{j-1} \right) \cdot \mathbb{1}_{\{0 \leq S \leq M \cdot z\}}.$
- (2) Sample $\beta_j \sim \text{IG} \left(\alpha_{j-1} \left\{ m\tilde{t}_{e,1} + \sum_{k=1}^M t_{n_k} \right\}, \left\{ m\tilde{z}_e + \tilde{S}_j + \sum_{k=1}^M \sum_{i=r_k+1}^{n_k} z_{k,i} \right\} \right).$
- (3) Sample $\tilde{\alpha}_j \sim \rho(\cdot|\alpha_{j-1}) \equiv \mathcal{N}(\alpha_{j-1}, \rho \cdot \alpha_{j-1}^2).$
- (4) Sample $u_j \sim \mathcal{U}[0, 1]$ and accept $\alpha_j = \tilde{\alpha}_j$ if $u_j \leq \eta_j$ where

$$\eta_j = \min \left\{ 1 \wedge \frac{\ell(\beta_j, \tilde{\alpha}_j|\mathbf{d}) \pi(\beta_j, \tilde{\alpha}_j) \rho(\alpha_{j-1}|\tilde{\alpha}_j)}{\ell(\beta_j, \alpha_{j-1}|\mathbf{d}) \pi(\beta_j, \alpha_{j-1}) \rho(\tilde{\alpha}_j|\alpha_{j-1})} \right\}$$

else accept $\alpha_j = \alpha_{j-1}.$

4. Maintenance decision rules

Once the deterioration parameters are estimated, the failure can be predicted, and an appropriate maintenance strategy can be proposed. As mentioned earlier, the cracks are not continuously monitored, and the data are only collected through inspections. The time of a preventive maintenance action has to be carefully chosen before the deterioration level exceeds the threshold μ . But if the replacement occurs too early, then the maintenance policy is conservative, and the lifecycle length can artificially be reduced. In the opposite, if the replacement takes place too late, the system can be failed.

Denote T_μ the failure time, namely the first hitting time of the threshold μ :

$$T_\mu = \inf\{t > 0, X_t > \mu\},$$

the cumulative (predictive posterior) distribution function of which being

$$\mathbb{P}(T_\mu \leq t) = \iint \mathbb{P}(X_t \geq \mu | \alpha, \beta) \pi(\alpha, \beta | \mathbf{d}) d\alpha d\beta = \iint \frac{\Gamma(\alpha t, \mu/\beta)}{\Gamma(\alpha t)} \pi(\alpha, \beta | \mathbf{d}) d\alpha d\beta.$$

For a given loss function L , the preventive replacement time is defined as the so-called Bayes estimator

$$\hat{T}_\mu = \operatorname{argmin}_x \iint \int L(x, t) \frac{\partial \Gamma(\alpha t, \mu/\beta)/\Gamma(\alpha t)}{\partial t} \pi(\alpha, \beta | \mathbf{d}) d\alpha d\beta dt \quad (10)$$

where, according to Park and Padgett [57]:

$$\begin{aligned} \forall t \geq 0, \quad \frac{\partial \Gamma(\alpha t, \mu/\beta)/\Gamma(\alpha t)}{\partial t} &= \alpha \left(\Psi(\alpha t) - \log\left(\frac{\mu}{\beta}\right) \right) \frac{\gamma(\alpha t, \mu/\beta)}{\Gamma(\alpha t)} \\ &+ \frac{1}{\alpha t^2 \Gamma(\alpha t)} \left(\frac{\mu}{\beta}\right)^{\alpha t} {}_2F_2(\alpha t, \alpha t; \alpha t + 1, \alpha t + 1; -F/\beta) \end{aligned}$$

is the sampling density of T_μ , with Ψ the digamma function and ${}_2F_2$ the generalized hypergeometric function of order (2, 2). An inversion method can be numerically used to sample from this density, given a couple (α, β) . Therefore, the estimation of any function of interest of T_μ can be computed by Monte Carlo means, mixing posterior and conditional samplings. It must be reminded that the posterior predictive integration (10), which underlies the Bayes estimator $\hat{T}_\mu$, by accounting for all parametric uncertainties expressed through the posterior distribution, induces a total order rule over the range of all possible estimators of T_μ [34]. Therefore, it is appropriate to define $\hat{T}_\mu$ as the minimizer of an integrated loss function L to get a coherent decision.

4.1. Different choices of loss function

The choice of the loss function is therefore of great importance to orient the maintenance strategy. Different choices of L will lead to a range of these strategies, from a more conservative one to a risk-averted one. Nonetheless, determining the most accurate loss functions is usually not possible for reliability engineers who cannot integrate the knowledge of all costs and benefits of maintenance actions within decision-theoretic frameworks at the level of (let us say) a full production plant. Therefore, simple choices must be made in practice. Three different but simple loss functions are considered in this purpose. Their difference is mainly due to the way the replacements are penalized and the failure is dealt with. What allows to select one among them is mainly the fact that a maintenance task can, in some situations, be operated on one components only or several components during the same visit (namely, without significant additive costs).

- The usual quadratic loss function $L_1(x, t) = ||x - t||^2$ penalizes more the large deviations from the real failure time. Therefore, this function is suitable if it is required to avoid too early replacements and is admitted to pay the price of possible late replacements. In this case, according to classical Bayesian estimation results [34], the replacement time $\hat{T}_\mu$ is the mean of the posterior distribution defined as follows:

$$\hat{T}_\mu = \iint \int t \frac{\partial \Gamma(\alpha t, \mu/\beta)/\Gamma(\alpha t)}{\partial t} \pi(\alpha, \beta | \mathbf{d}) d\alpha d\beta dt$$

which can be computed by Monte Carlo means, as explained earlier. If $\beta\mu$ is large enough, the conditional expectation of T_μ , $\mathbb{E}(T_\mu)$, is approximately equal to $\frac{1}{\alpha}(\beta\mu + \frac{1}{2})$ (cf. [58]), therefore

$$\hat{T}_\mu \simeq \iint \frac{1}{\alpha} \left(\beta\mu + \frac{1}{2} \right) \pi(\alpha, \beta | \mathbf{d}) d\alpha d\beta.$$

Because the quadratic loss function appears as the second-order Taylor-type expansion of any subjective loss function, this Bayes estimator is certainly the most used in practice.

- However, the absolute error loss function $L_2(x, t) = |x - t|$ has also interest since it treats similarly the large and small deviations from the real failure time. In this case, the corrective and preventive maintenance are similarly handled, which means that the failure is not necessarily more costly than the preventive replacement. This loss function is therefore more suited to systems where the maintenance of any kind (preventive or corrective) is costly, and there is no advantage in early preventive maintenance actions. This is often the case with a unique component, the impact of which on the system safety being low. In this case, the replacement time $\hat{T}_\mu$ is calculated as follows:

$$\frac{\partial}{\partial x} \iiint |x - t| \frac{\partial \Gamma(\alpha t, \mu/\beta)/\Gamma(\alpha t)}{\partial t} \pi(\alpha, \beta | \mathbf{d}) dt d\alpha d\beta = 0$$

which implies $\mathbb{P}(T_\mu \leq \hat{T}_\mu) = \frac{1}{2}$. Hence, the best replacement time $\hat{T}_\mu$ is the median of the marginal posterior distribution of T_μ .

- Finally, the linear loss function $L_3(x, t) = C_1|x - t|\mathbb{1}_{x \geq t} + C_2|x - t|\mathbb{1}_{x \leq t}$ is adapted to cases where the late replacements are penalized in the sense that the failure (exceeding of the threshold μ) incurs a cost C_1 and the preventive replacement incurs a cost C_2 with $C_2 \leq C_1$. It is highly incited to replace the system before failure and to avoid a resulting period of unacceptable activity. In this case the preventive replacement time is the solution of the following equation:

$$\frac{\partial}{\partial x} \iiint L_3(x, t) \frac{\partial \Gamma(\alpha t, \mu/\beta)/\Gamma(\alpha t)}{\partial t} \pi(\alpha, \beta | \mathbf{d}) dt d\alpha d\beta = 0$$

which leads to

$$\mathbb{P}(T_\mu \leq \hat{T}) = \frac{C_1}{C_1 + C_2}.$$

Hence, the best replacement time $\hat{T}_\mu$ is the $C_1/C_1 + C_2$ -order quantile of the marginal predictive posterior distribution of T_μ , which can still empirically computed from a Monte Carlo sampling from this latter distribution. Denoting $C_1 = \delta C_2$, where $\delta \in \mathbb{R}^{+*}$, the replacement time $\hat{T}_\mu$ is the $\delta/(1 + \delta)$ -order quantile of T_μ . In this framework, as δ grows, the corrective actions become very expensive and the time before a preventive replacement becomes, logically, very short.

Considering that the corrective action is on the whole system (all the components) and the preventive actions are only carried out on inefficient components, it can be assumed, in a first approximation, that $C_1 \simeq QC_2$ where Q is the number of components under consideration. Because there is no need to specify absolute value for the cost, this makes this loss function particularly attractive in the industrial context of the French plant park, where the maintenance visits have multiple costs that are mainly driven by time.

5. Application to real data

An EDF dataset, providing the sizes of $M = 26$ transverse cracks appearing on rotor supports at random isolated times, was collected between 1977 and 2012 over 15 pressurized water reactors belonging to the French park. These rotors take part in the processes of steam cooling of the secondary water circuit of each reactor. During its lifetime, each component is inspected 7.7×, the total number of available inspections being 200, and a new inspection occurs on average 3.26 years after the previous one. These in-service observations are scarce because they are provided using an ultrasonic measurement process, which requires rather costly technical and human means, and furthermore the technological evolution of this process prevents to make use of other (e.g., old) data. The accuracy of the measurement process was ensured for a threshold value of $z = 18$ mm (100% of probability for a crack to be detected). This resulted in a mean censoring percentage of 68% per component sample. The distribution of truly observed (uncensored) data is displayed in Figure 1, and the observed trajectories are displayed in Figure 2.

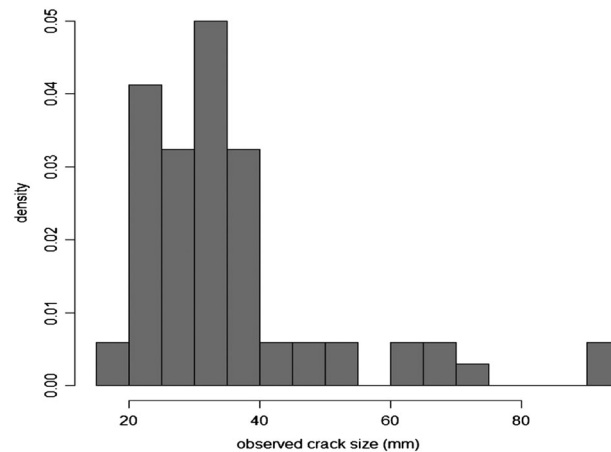


Figure 1. Histogram of truly observed crack sizes. All observations are above the detection threshold $z = 18\text{mm}$.

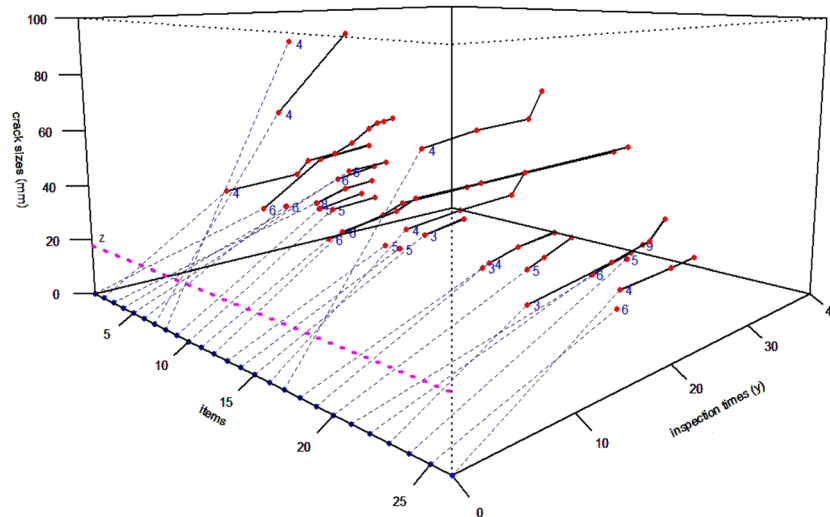


Figure 2. Observed trajectories of crack size over 26 similar components. The dashed blue lines correspond to censored parts of the trajectories, and the numbers of missing data per observed trajectory are written in blue.

Considering only the components on which at least one crack appeared, former studies on close ones have shown that the choice of a gamma process is not irrelevant for modeling the crack increasing [21, 59]. However, testing formally the validity of this assumption remains an open issue. Indeed, to the current state of our knowledge, very few studies were conducted about developing graphical preliminary analyses (such as equivalents of Weibull plots) or goodness-of-fit tests in the case of gamma processes. Dedicated likelihood ratio [60] and Kolmogorov-Smirnov [61] tests were proposed only in situations where the data are assumed to be perfect observations (e.g., without censoring) and, for the latter case, collected in asymptotic settings. Clearly these approaches cannot be used in the present case. Using the left-censored Kaplan-Meier non-parametric estimator of the cumulative distribution function (CDF) F_X , in view of graphical comparisons with a fitted DCF arising from a gamma assumption, could appear as a more appropriate approach. But the obvious dependance of the $X_{k,t}$ along the trajectory k does not allow to establish an unbiased estimator of F_X . An alternative approach could be to define a Kaplan-Meier estimator of the CDF of the $Z_{k,t}$, which are assumed to be independent. Unfortunately, this estimator would be doubly censored and consequently not consistent. Finally, the relevance of the fitted model is tested further using posterior predictions (as usual in the Bayesian community).

Independently of collecting the data, an EDF technical expert who worked on several pressurized water reactor plants was questioned about the main features of the degradation process given a crack has appeared. He expressed the qualitative opinion that the degradation process could not be accelerated, because of the passive nature of the components, the way that they are used in running conditions, and the toughness properties of their materials. Gamma processes are likely to model such a behavior, because the deterioration speed rate is linear in time.

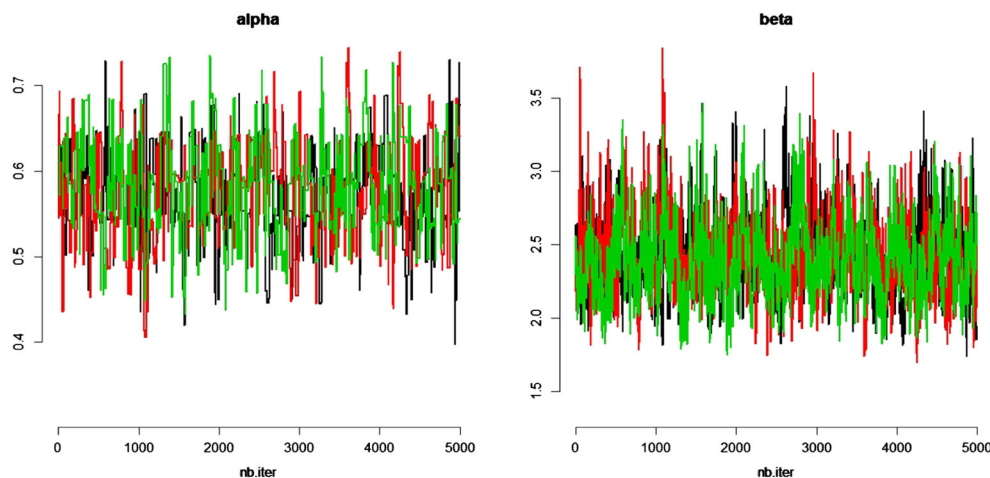


Figure 3. Three parallel Gibbs chains sampling within the posterior distribution of (α, β) , around the maximum likelihood estimator $(\hat{\alpha}, \hat{\beta}) = (0.6, 2.31)$. The prior distribution is indeed chosen noninformative.

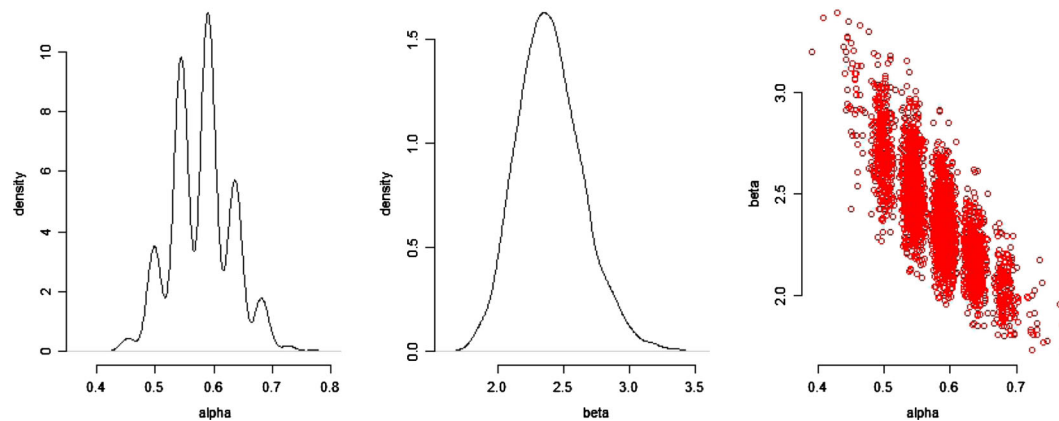


Figure 4. Marginal posterior distributions of (α, β) and joint posterior sampling, given a non-informative prior.

A first estimation study was conducted using maximum likelihood (ML) maximization and the objective Bayesian approach, based on a burn-in period of 2000 iterations. Sections of the corresponding Markov chains at stochastic equilibrium are plotted in Figure 3. As it could be expected, the MCMC chains concentrate around the ML estimator (MLE) of value $(\hat{\alpha}, \hat{\beta}) = (0.6, 2.31)$ because no truly informative prior knowledge is provided. The wide range of posterior uncertainty that can be noticed on Figure 3 limits the relevance of using the best estimate for the yearly deterioration speed rate $r_{\hat{\alpha}, \hat{\beta}}(1) = \hat{\alpha}\hat{\beta} = 1.386$ mm per year. It must be furthermore noticed that the likelihood suffers from multimodality (in the parametric space where α lives), as reflected by its posterior distribution (Figure 4). This unexpected phenomenon is due to the high censoring affecting the likelihood. It highlights the practical difficulties raised by choosing a single point estimate of (α, β) , as the MLE, to summarize the behavior of the lifetime of any similar component and the need for taking into account the parametric uncertainties coherently (namely, in a probabilistic way), which are expressed by the posterior distribution in the Bayesian framework. Despite those difficulties, the global relevance of the estimated gamma process can be perceived in Figure 5 where samples of the objective posterior distribution are projected in terms of predicted crack size through time. The posterior predictive p -values corresponding to the location of each observed data remain for nearly 90% of them in the range 25–75%.

Nonetheless, the obtained value of 1.386 mm per year for the deterioration rate was considered slightly underestimated by the expert, which fits with the fact that the highest observed data remain in the upper tail of the predicted posterior distribution (cf. Figure 5). Instead, he advocated a speed of 2 mm per year as the most likely crack increasing. This is interpreted as the modal value of $r(1)$, namely $\tilde{z}_e = 2\tilde{t}_{e,1}$ from (8). Coherent with this information, the full calibration of hyperparameters depends on the ability of the expert to forecast how the cracks will develop over the years. Actually, the expert was trained to basic statistical knowledge, as recommended in [62] and [52], then questioned as follows: during the next 15 then 30 years (i.e., the value of $m\tilde{t}_{e,1}$), what will be the chances $(1 - \delta_1, 1 - \delta_2)$ that any crack of size X appearing

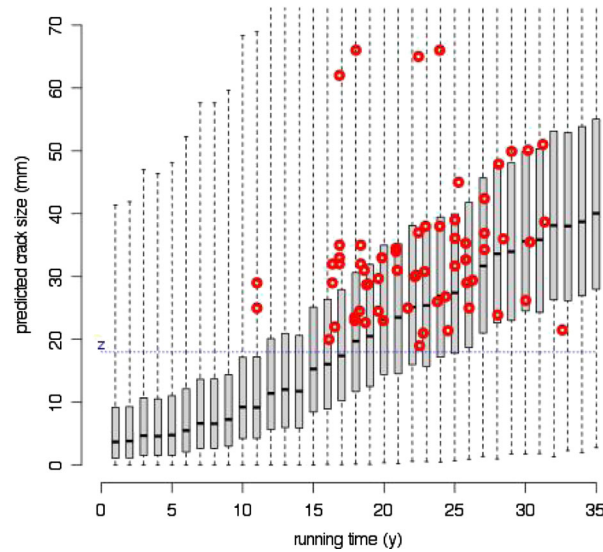


Figure 5. Boxplots of the posterior distributions (given an non-informative prior on (α, β)) of each predicted crack size and real observations (in red). The boxes represent the interquartile intervals (25-75%-ordered percentiles) around the median, and the lines indicate the intervals 1–99%. 5000 posterior samples produced by the Gibbs algorithm are used to compute these box plots.

on the device be upper than $(x_1, x_2) = (5, 10)$ mm? In statistical terms, it means that for $i = \{1, 2\}$,

$$\begin{aligned} P\left(X_{m\tilde{t}_{e,1}} < x_i\right) &= \delta_i, \\ &= \int_0^{x_i} \int_0^\infty \int_0^\infty \frac{\beta^{-\alpha m\tilde{t}_{e,1}}}{\Gamma(\alpha m\tilde{t}_{e,1})} x^{\alpha m\tilde{t}_{e,1}-1} \exp(-x/\beta) \pi(\beta|\alpha) \pi(\alpha) d\beta dx d\alpha, \\ &= \int_0^{x_i} \int_0^\infty \frac{x^{\alpha m\tilde{t}_{e,1}-1} (m\tilde{z}_e)^{\alpha m\tilde{t}_{e,1}} \Gamma(2\alpha m\tilde{t}_{e,1})}{(m\tilde{z}_e + x)^{2\alpha m\tilde{t}_{e,1}} \Gamma^2(\alpha m\tilde{t}_{e,1})} \pi(\alpha) d\alpha dx. \end{aligned} \quad (11)$$

In each situation, the expert provided a couple of possible values for δ_i in the range $\{1\%, 5\%, 10\%, 20\%, 25\%\}$ provided by the Bayesian analyst, corresponding to (crude) optimist and pessimist cases. Both integral equations (11) can be numerically solved in $(m, \tilde{t}_{e,2})$ using a Monte Carlo approach computed over a bidimensional grid. The criterion to minimize is chosen as a relative least-square regret:

$$C(m, \tilde{t}_{e,2}) = \sum_{i=1}^2 \delta_i^{-2} \left\{ \int_0^{x_i} \int_0^\infty \frac{x^{\alpha T-1} (2T)^{\alpha T} \Gamma(2\alpha T)}{(2T + x)^{2\alpha T} \Gamma^2(\alpha T)} \pi(\alpha) d\alpha dx - \delta_i \right\}^2$$

where $T \in \{15, 30\}$ years. All optimization tasks led to a unique singleton $(m, \tilde{t}_{e,2})$. Hence, several priors (indexed by $(a), \dots, (d)$) can be elicited, the features of which being summarized on Table I, that cover the range of plausible values for (α, β) according to the optimist and pessimist views of the expert. The differences between the expected and effective percentile orders are due to the stiffness of the prior distribution, which cannot fit exactly the requirements arising from the expert's opinion (such a fit can never be found in reality, as stressed in [62]). Instead, the priors summarized on Table I must be seen as the best possible trade-offs between the chosen prior structure (coherent with the sampling model) and the opinion of the expert, in the L_2 sense. It is interesting to notice that the prior elicitation gives more weight (i.e., a larger virtual size) to the optimist views of the expert. Nonetheless, in each case, the posterior remains mainly driven by the observed data information, rather than by the subjective opinion.

In this application, two quantities of interest must be estimated. First, the lifetime T_{40} until reaching theoretically a threshold crack size $\mu_1 = 40$ mm is useful to predict when yearly visits of installations should start. Such visits are mandatory, with a given cost to put aside, once this size has been reached and observed after a visit. The posterior distributions of this visiting time are plotted over Figure 6, corresponding to the range of priors previously elicited (40,000 Gibbs iterations were used). It must be noticed that the impact of the prior makes the posterior wider than in the non-informative case, which is coherent with the cautiousness expressed by the expert; in substance, it means that the dataset only does not testify of all the uncertainty that surrounds the distribution of the visiting (then failure) time. Because the strongest gap between the optimal and the chosen visiting times must be penalized, an L_2 quadratic loss function is used to define the

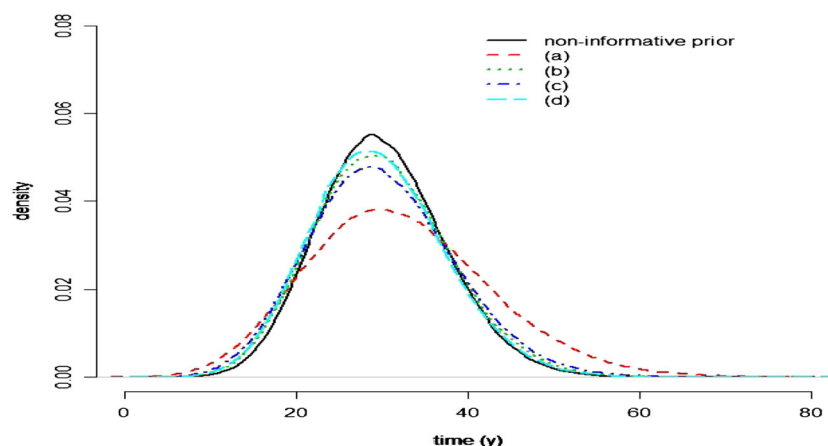


Figure 6. Posterior predictive density functions of the visiting time T_{40} given noninformative and informative priors.

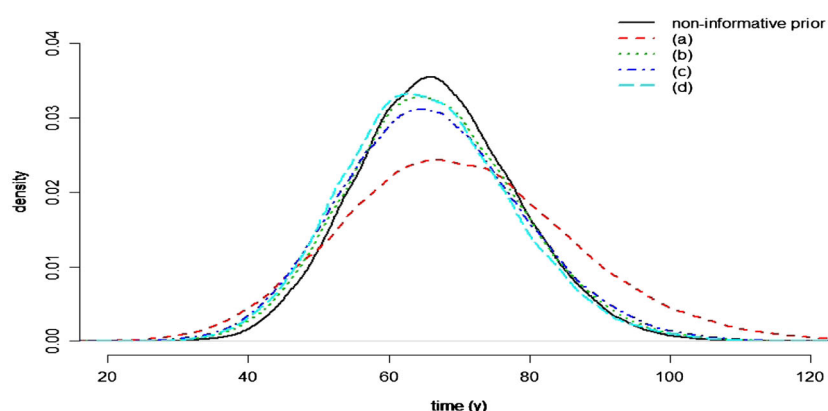


Figure 7. Posterior predictive density functions of the failure time T_{90} given noninformative and informative priors.

Table I. Features of prior elicitation for various situations (*optim.* and *pessim.* means *optimist* and *pessimist*) of crack increasing, proposed by the industrial expert.

	Forecasting time			
	15 years 30		30 years 60	
Period of virtual observations $m\tilde{t}_{e,1}$				
Most probable virtual crack size $m\tilde{z}_e$ (mm)	opt.	pess.	opt.	pess.
Expected chances that $X < 5$ mm	10%	5%	1%	5%
Expected chances that $X < 10$ mm	25%	15%	5%	10%
Calibrated m	21	6	6	2.25
Calibrated $\tilde{z}_e$	1.43	5	10	26.7
Calibrated $\tilde{t}_{e,1}$	0.71	2.5	5	13.3
Calibrated $\tilde{t}_{e,2}$	9.25	5	9.3	7.8
Effective chances that $X < 5$ mm	10.8%	5.5%	1.3%	5.2%
Effective chances that $X < 10$ mm	23.6%	14.5%	4.9%	10.4%
Prior index	(a)	(b)	(c)	(d)

Note: The expected chances are provided by the expert and the effective chances result from sampling the prior predictive distribution of X based on the numerical calibration of $(m, \tilde{t}_{e,2})$ on a grid in the bounded domain $[1, 30] \times [0.001, 20]$ by a least square rule.

estimator $\hat{T}_{40}$. By averaging on the posterior distributions (in absence of clear rule allowing to discriminate them), it was found $\hat{T}_{40} = 30.4$ years. Furthermore, in a French production system, the considered components are usually two, running in parallel. Therefore, following the ideas expressed in the previous section, it seems reasonable to choose the 33%-order percentile of the failure time distribution (plotted in Figure 7) as the best time $\hat{T}_{rep}$ to proceed to preventive maintenance. The considered size μ defining a so-called 'failure' is 90 mm. The most cautious assessment indicates a best maintenance time of 58 years, which differs from the 61 years proposed by the non-informative posterior modeling.

6. Discussion

This article provides a comprehensive Bayesian approach to estimate a gamma degradation process and to compute maintenance times by taking account of parametrical uncertainty, in classical situations where a large part of the data is censored due to the limits of the experimental process. Based on representativeness heuristics [63] underlying the expression of the prior opinion, this methodology is adapted to cases where some knowledge is possibly available to supplement the data information. The applicability of the methodology is demonstrated on a real industrial case study.

Of course, this approach should be interpreted only as a basis for more detailed elicitation methodologies, involving for instance hyperprior uncertainties around the estimates provided by the expert and the virtual size. Indeed, in this first application, the opinion of the expert is fully integrated without questioning the possible cognitive and motivational bias that can affect his or her judgment [52]. Many extensions can be provided, simplified by interpreting the sense of all involved hyperparameters. For instance, to ensure posterior objectivity, one could follow the choice suggested in [64], in the particular field of Bayesian linear regression, of eliciting a non-informative (hyper)prior for the virtual size. However, in general, our opinion is that such extensions intrinsically depend on the case study. Another interest of the proposed Bayesian model is that it can be used as a relevant benchmark prior to which any other informative prior may be compared, to compute the underlying amount of subjective information (for instance by minimizing some distance or divergence in the Ali-Silvey class).

Some points should deserve a deeper study, the interest of which having not appeared essential within the scope of the present paper. The elicitation of a Jeffrey's prior that would take account of the censoring phenomena affecting the likelihood and the Fisher matrix is of interest, as [65] showed on simple examples that such priors can improve the properties of the posterior distribution in terms of frequentist coverage when dealing with small samples. This approach was not carried out here because it requires laborious technical developments using incomplete polygamma functionals, for a result which was suspected to present low differences in terms of maintenance management.

In future studies, a welcomed extension of this work would be to include in the Bayesian modeling and decisional framework the misknowledge of crack initiations, which appear at random times, for instance using a Poisson process (as proposed in [21, 59]). This will complete the overall treatment of uncertainties appearing during the data collection when considering identical systems running independently in similar conditions. To go further, in order to account of possible correlations due to running conditions that make cracks appear during parallel degradation processes, the statistical likelihood should have to be modified. Two or more correlated gamma processes would share prior features on their parameters, for instance, following a hierarchical approach [34]. This would require to adapt the methodology proposed in this article. Following alternatively the so-called supra-Bayes approach originally proposed in [66], the observed degradations could be fully considered as correlated samples from an overall gamma process modeling a hidden (virtual) degradation. In this situation, the principles of prior elicitation proposed in this article can still be carried out.

Finally, the possible use of several processes to model the behavior of any positive quantity (as a crack) is becoming a challenge in the field of reliability and risk assessment. Various tools as Wiener processes with positive drift, power-law processes or semi-parametric gamma processes can be estimated and should be compared with the light of the observed data and through their respective predictive power. Both objectives require to develop comparable assessment methods and to make use of specific comparison statistics. First, efforts should be focused on developing goodness-of-fit tests adapted to small trajectory samples submitted to a high level of left censoring. Such features are indeed common in industrial reliability studies where costly experiments are made using indirect measurement means (e.g., ultrasonic or X-ray devices) to follow through time a process of deterioration [67]. Most often, the censoring limit z can be replaced by a so-called probability of detection characterizing the measurement mean, which is itself stamped uncertainty because it is calibrated from a small number of experiments conducted on mock-ups [68]. Composite likelihoods recently proposed by Mercier *et al.* [59] seem promising tools to develop such methodologies of testing, despite the strong dependency between the observations of trajectory random variables.

Second, in the Bayesian framework, the traditional use of Bayes factors would need to develop prior elicitation methods, which remain 'fair' in the sense that they do not favor a more complex model with respect to a simpler one [69]. Since the strength of the prior information is here elicited as a virtual size given the gamma sampling model, it must be expected

that the calibration of this size depends on the complexity of any other sampling model considered in a selection process. It appears essential, in a future work, to develop such methodologies of prior compatibility, as recommended in [64, 69].

APPENDIX A. RESULTS OF OBJECTIVE BAYESIAN EXPERIMENTS

Table A.1. Summary of objective posterior computations averaged on replicated datasets simulated using $\alpha = 5$ and $\beta = 0.1$.

n	M	z	Censor. rate	Bias α	MSE α	Bias β	MSE β
20.00	20.00	6.00	24.50	17.88	21.76	-16.10	0.32
		12.00	46.90	77.42	318.68	-44.70	2.03
		18.00	70.30	159.48	1311.75	-62.60	3.93
	40.00	6.00	25.00	-3.69	3.17	4.49	0.09
		12.00	48.20	37.75	78.33	-29.20	0.89
		18.00	70.00	54.18	153.24	-37.91	1.46
	80.00	6.00	25.00	-47.20	112.10	92.25	8.71
		12.00	46.30	-10.80	7.81	9.12	0.14
		18.00	70.00	-3.08	2.86	-2.32	0.05
40.00	20.00	6.00	23.15	1.46	1.95	-3.45	0.05
		12.00	47.00	4.01	3.72	-7.12	0.10
		18.00	71.10	24.36	50.20	-22.61	0.70
	40.00	6.00	22.90	2.34	1.75	-4.40	0.05
		12.00	47.50	0.22	2.62	-4.04	0.07
		18.00	70.95	15.65	22.45	-18.07	0.43
	80.00	6.00	22.95	2.01	0.65	-4.37	0.03
		12.00	47.05	1.26	1.54	-5.27	0.06
		18.00	70.70	14.36	16.94	-17.80	0.39
80.00	20.00	6.00	23.05	-0.59	0.82	-2.09	0.03
		12.00	47.17	-5.96	2.62	2.56	0.04
		18.00	71.60	-9.37	7.28	4.40	0.12
	40.00	6.00	22.90	-2.20	0.59	-0.10	0.01
		12.00	47.02	-5.50	2.47	1.79	0.03
		18.00	71.55	-9.39	5.78	3.49	0.05
	80.00	6.00	23.00	-2.59	0.52	-0.12	0.00
		12.00	47.05	-6.13	2.05	2.10	0.01
		18.00	71.50	-9.72	5.36	3.58	0.04

MSE, mean square error.

Table A.2. Summary of objective posterior computations averaged on replicated datasets simulated using $\alpha = 5$ and $\beta = 0.2$.

n	M	z	Censor. rate	Bias α	MSE α	Bias β	MSE β
20.00	20.00	6.00	11.50	1.50	2.06	-0.24	0.09
		12.00	24.60	21.59	26.97	-18.49	0.76
		18.00	35.00	43.66	105.18	-31.29	2.05
	40.00	6.00	11.20	-50.16	127.64	116.25	28.30
		12.00	25.00	-3.93	4.05	4.59	0.22
		18.00	35.00	15.60	15.19	-14.48	0.48
	80.00	6.00	10.90	-75.91	288.17	361.65	262.46
		12.00	25.00	-47.22	112.23	92.73	17.64
		18.00	35.00	-21.18	23.92	25.55	1.47

Table A.2. Continued.

n	M	z	Censor. rate	Bias α	MSE α	Bias β	MSE β
40.00	20.00	6.00	11.50	2.57	2.03	-3.54	0.09
		12.00	23.45	2.54	3.24	-3.79	0.14
		18.00	35.00	2.52	4.30	-4.77	0.19
	40.00	6.00	11.20	2.87	1.08	-4.15	0.06
		12.00	23.00	4.15	2.01	-6.23	0.11
		18.00	35.35	2.42	1.79	-5.08	0.11
	80.00	6.00	11.35	1.19	0.43	-2.64	0.03
		12.00	22.95	2.28	0.75	-4.43	0.06
		18.00	35.00	2.84	1.28	-6.06	0.11
	20.00	6.00	10.90	-0.59	0.84	-0.71	0.04
		12.00	22.88	-2.94	1.01	0.69	0.03
		18.00	34.73	-4.44	1.48	2.06	0.03
80.00	40.00	6.00	10.97	-1.28	0.28	-0.65	0.01
		12.00	23.05	-2.06	0.45	0.18	0.01
		18.00	34.98	-4.70	1.32	1.86	0.02
	80.00	6.00	10.93	-0.86	0.19	-0.73	0.01
		12.00	23.00	-2.53	0.47	0.25	0.01
		18.00	35.10	-3.48	0.86	0.39	0.01

MSE, mean square error.

Table A.3. Summary objective posterior computations averaged on replicated datasets simulated using $\alpha = 10$ and $\beta = 0.1$.

n	M	z	Censor. rate	Bias α	MSE α	Bias β	MSE β
20.00	20.00	6.00	11.50	-46.27	216.80	97.55	9.93
		12.00	25.00	-23.49	59.20	33.37	1.21
		18.00	35.00	-5.73	12.96	6.83	0.16
	40.00	6.00	11.10	-79.81	637.25	441.75	197.91
		12.00	25.00	-60.81	370.60	161.60	26.47
		18.00	35.00	-44.98	203.68	82.49	6.96
	80.00	6.00	10.60	-88.17	777.47	860.75	742.01
		12.00	25.00	-75.37	568.07	319.50	102.25
		18.00	35.00	-65.13	424.51	188.39	35.71
	20.00	6.00	11.10	5.33	6.83	-5.88	0.07
		12.00	22.90	11.03	20.93	-10.36	0.17
		18.00	34.95	18.36	40.64	-17.09	0.33
40.00	40.00	6.00	11.55	2.17	1.28	-2.41	0.01
		12.00	23.00	8.92	10.04	-9.78	0.11
		18.00	35.00	17.95	37.97	-16.82	0.31
	80.00	6.00	11.05	-37.39	145.85	66.57	4.79
		12.00	22.60	6.18	4.47	-7.00	0.05
		18.00	35.00	11.16	14.03	-12.06	0.15
	20.00	6.00	10.78	-0.45	0.64	-0.36	0.01
		12.00	22.98	-2.23	1.96	0.38	0.01
		18.00	34.75	-2.86	2.38	1.26	0.02
	80.00	6.00	10.75	1.16	0.49	-2.38	0.01
		12.00	22.82	-1.25	0.72	-0.41	0.01
		18.00	34.98	-2.92	1.86	0.76	0.01
80.00	80.00	6.00	11.10	0.30	0.32	-1.44	0.01
		12.00	22.75	-0.85	0.63	-0.70	0.01
		18.00	34.85	-3.30	1.64	1.04	0.01

MSE, mean square error.

Table A.4. Summary of objective posterior computations averaged on replicated datasets simulated using $\alpha = 10$ and $\beta = 0.2$.

n	M	z	Censor. rate	Bias α	MSE α	Bias β	MSE β
20.00	20.00	6.00	5.00	-70.71	501.60	282.45	165.03
		12.00	11.30	-47.50	229.81	103.57	23.13
		18.00	19.80	-32.05	104.89	51.32	5.49
	40.00	6.00	5.00	-82.68	683.63	532.98	568.74
		12.00	11.20	-78.68	619.32	405.34	331.96
		18.00	19.90	-68.18	465.38	227.96	104.98
	80.00	6.00	5.00	-82.63	682.79	532.00	566.90
		12.00	10.20	-82.73	684.47	527.22	556.46
		18.00	19.90	-81.18	659.13	465.24	433.66
40.00	20.00	6.00	5.00	2.19	2.54	-2.16	0.05
		12.00	10.95	5.02	5.85	-5.18	0.11
		18.00	17.45	8.02	11.79	-8.25	0.23
	40.00	6.00	5.00	-33.74	125.94	74.38	12.88
		12.00	11.40	2.80	2.37	-2.92	0.05
		18.00	17.50	6.23	6.06	-6.96	0.13
	80.00	6.00	5.00	-76.16	580.23	382.33	293.94
		12.00	11.35	-37.26	142.26	65.57	9.06
		18.00	17.50	-2.21	3.14	1.98	0.08
	20.00	6.00	4.92	1.46	1.49	-1.73	0.03
		12.00	10.73	0.32	1.07	-1.81	0.03
		18.00	16.68	-0.74	1.72	-0.47	0.03
	40.00	6.00	4.98	1.44	0.44	-2.18	0.01
		12.00	10.95	0.68	1.09	-1.57	0.02
		18.00	16.70	-0.02	1.15	-1.20	0.03
	80.00	6.00	5.00	0.52	0.39	-1.32	0.01
		12.00	11.03	0.59	0.27	-1.67	0.01
		18.00	16.70	-0.58	0.34	-0.94	0.01

MSE, mean square error.

APPENDIX B. PROOF OF PROPOSITION 1

Denote $a = m/2$, $b = m\tilde{m}_{e,2}$, $c = m\tilde{m}_{e,1}$ and $d = m\tilde{z}_e$. Using (5) and (6), the prior predictive cumulative function of $r(\Delta_i) = \alpha\beta\Delta_i$ is, $\forall x \geq 0$,

$$\begin{aligned}
 P(r(\Delta_i) \leq x) &= \int_0^\infty P\left(\beta \leq \frac{t}{\alpha\Delta_i}\right) \pi(\alpha) d\alpha, \\
 &= \int_0^\infty \int_0^{\frac{x}{\alpha\Delta_i}} \frac{d^{\alpha c}}{\Gamma(\alpha c)} \beta^{-\alpha c-1} \exp(-d/\beta) \pi(\alpha) d\beta d\alpha, \\
 &= \int_0^\infty \int_0^x \frac{d^{\alpha c}}{\Gamma(\alpha c)} \left(\frac{y}{\alpha\Delta_i}\right)^{-\alpha c-1} \exp(-d\alpha\Delta_i/y) \pi(\alpha) \frac{dy}{\alpha\Delta_i} d\alpha, \\
 &= \int_0^x \left[\int_0^\infty \frac{y^{-1}b^a}{\Gamma(a)\Gamma(\alpha c)} \alpha^{ac+a-1} \exp\left(-\alpha\left\{b + \frac{d\Delta_i}{y} - c \log \frac{y}{d}\right\}\right) d\alpha \right] dy.
 \end{aligned}$$

Therefore, the prior predictive probability density function of $r(\Delta_i)$ is

$$f(x) = \int_0^\infty \frac{x^{-1}b^a}{\Gamma(a)\Gamma(\alpha c)} \alpha^{ac+a-1} \exp\left(-\alpha\left\{b + \frac{d\Delta_i}{x} - c \log \frac{x}{d}\right\}\right) d\alpha$$

which is maximal in $x^* = \hat{r}(\Delta_i)$ defined by

$$\begin{aligned} f'(x^*) &= 0, \\ &= C(x^*) \left[-\frac{d\Delta_i}{x^{*2}} + \frac{c}{x} \right] \end{aligned}$$

where

$$C(x) = \int_0^\infty \frac{x^{-1}b^a}{\Gamma(a)\Gamma(ac)} \alpha^{ac+a} \exp \left(-\alpha \left\{ b + \frac{d\Delta_i}{x} - c \log \frac{x}{d} \right\} \right) d\alpha$$

which is strictly positive for $x > 0$. Therefore $x^* = d\Delta_i/c = \tilde{z}_e \Delta_i / \tilde{t}_{e,1}$.

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References

- Barlow R, Proschan F. Statistical Theory of Reliability and Life Testing. *International Series in Decision Processes ; Edition Holt*. Rinehart and Winston: New York, 1975.
- Barlow R, Proschan F. Mathematical Theory of Reliability. *Society for Industrial and Applied Mathematics ; Classics in Applied Mathematics*. John Wiley and Sons: New York, 1996.
- Newby MJ. Perspective on Weibull proportional-hazards model. *IEEE Transactions on Reliability* 1994; **43**:217–223. DOI: 10.1109/24.294993.
- Barker C, Newby M. Optimal non-periodic inspection for multivariate degradation model. *Reliability Engineering and System Safety* 2009; **94**:33–43. DOI: 10.1016/j.ress.2007.03.015.
- Nicola RP, Dekker R, van Noortwijk JM. Comparison of models for measurable deterioration : an application to coatings on steel structures. *Reliability Engineering and System Safety* 2007; **92**:1635–1650. DOI: 10.1016/j.ress.2006.09.021.
- Ross S. Multivaluated state component systems. *Annals of Probability* 1979; **7**:379–383. DOI: 10.1214/aop/1176995096.
- Abdel-Hameed M, Proschan F. Non-stationary shock models. *Stochastic processes and Their Applications* 1973; **1**:383–404. DOI: 10.1016/0304-4149(73)90019-7.
- Nikulin MS, Limnios N, Balakrishnan N, Kahle W, Huber-Carol C. *Advances in Degradation Modeling : Applications to Reliability, Survival Analysis, and Finance*. Birkhäuser Basel: Boston, 2010.
- Singpurwalla ND. Survival in dynamic environments. *Statistical Science* 1995; **1**:86–103. DOI: 10.1214/ss/1177010132.
- Singpurwalla ND. Gamma processes and their generalizations: an overview. In *Engineering Probabilistic Design and Maintenance for Flood Protection*, Cooke, Mendel, Vrijling (eds), 1997, 67–73.
- Whitmore GA. Estimating degradation by a Wiener diffusion process subject to measurement error. *Lifetime Data Analysis* 1995; **1**:307–319. DOI: 10.1007/BF00985762.
- Abdel-Hameed M. A gamma wear process. *IEEE Transaction on Reliability* 1975; **24**:152–153. DOI: 10.1109/TR.1975.5215123.
- van Noortwijk JM, van Gelder P. Optimal Maintenance Decisions for Berm Breakwaters. *Structural Safety* 1996; **18**:293–309. DOI: 10.1016/0378-3839(92)90049-Z.
- van Noortwijk JM. A survey of the application of gamma processes in maintenance. *Reliability Engineering and System Safety* 2009; **94**:2–21. DOI: 10.1016/j.ress.2007.03.019.
- Dieulle L, Bérenguer C, Grall A, Roussignol M. Sequential condition-based maintenance scheduling for a deteriorating system. *European Journal of Operational Research* 2003; **150**:451–461. DOI: 10.1016/S0377-2217(02)00593-3.
- Grall A, Dieulle L, Berenguer C. Continuous - time predictive maintenance scheduling for a deteriorating system. *IEEE transactions on Reliability* 2002; **51**:141–150. DOI: 10.1109/TR.2002.1011518.
- Çinlar E, Bažant Z, Osman E. Stochastic process for extrapolating concrete creep. *Journal of the Engineering Mechanics Division* 1977; **103**:1069–1088. DOI: 10.1007/BF02474019.
- Cooke R, Mendel M, Vrijling H. *Engineering Probabilistic Design and Maintenance for Flood Protection*. Kluwer Academic Publishers: New York, 1997.
- Frangopol DM, Kallen MJ, van Noortwijk JM. Probabilistic models for life-cycle performance of deteriorating structures: review and future directions. *Progress in Structural Engineering and Materials* 2004; **6**:197–212. DOI: 10.1002/pse.180.
- Christer AH, Wang W, Sharp J. A state space condition monitoring model for furnace erosion prediction and replacement. *European Journal of Operational Research* 1997; **101**:1–14. DOI: 10.1016/S0377-2217(97)00132-X.
- Blain C, Barros A, Grall A, Lefebvre Y. Modelling of stress corrosion cracking with stochastic processes - application to steam generators. *Risk, Reliability and Societal Safety, Proceedings of the European Safety and Reliability Conference (ESREL)*, vol. 25–27, Stavanger, Norway, June 2007, 2395–2400.
- Lin D, Makis V. On-line parameter estimation for a failure-prone system subject to condition monitoring. *Journal of Applied Probability* 2004; **41**:211–220. DOI: 10.1239/jap/1077134679.

23. Bernardo JM, Smith AFM. *Bayesian Theory*. Wiley: New York, 1994.
24. Box G, Tiao G. *Bayesian Inference in Statistical Analysis*. John Wiley and Sons: New York, 1973.
25. Robert CP, Casella G. *Monte Carlo Statistical Methods*. Springer: New York, 2004.
26. Freitas MA, Colosimo EA, dos Santos TR, Pires MC. Reliability assessment using degradation models: Bayesian and classical approaches. *Pesquisa Operacional* 2010; **30**:194–219. DOI: 10.1590/S0101-74382010000100010.
27. Rios Insua D, Ruggeri F, Wiper MP. *Bayesian Analysis of Stochastic Proces Models*. Wiley: Chichester, 2012.
28. Guérin F, Barreau M, Demri A, Cloupet S, Hersant J, Hambli R. Bayesian Estimation of Degradation Model Defined by a Wiener Process. *Mathematical and Statistical Models and Methods in Reliability. Applications to Medicine, Finance and Quality Control*, VV Rikov, Balakrishnan N, Nikulin MS (eds). Birkhäuser, 2010.
29. Huibing H, Chun S. A Bayesian framework for reliability assessment via Wiener process and MCMC. *Mathematical Problems in Engineering* 2014, Article ID 486368:8 pages. DOI: 10.1155/2014/486368.
30. Ye Z, Chen N, Tsui KL. A Bayesian approach to condition monitoring with imperfect inspections. *Quality and Reliability Engineering International* 2013. DOI: 10.1002/qre.1609.
31. Zhou J. Bivariate degradation modeling based on gamma process. *Proceedings of the World Congress on Engineering 2010 Vol III WCE 2010*, London, UK, June 30–July 2, 2010.
32. Blanchard BS. *Logistics Engineering and Management (Fourth Ed.)*. Prentice-Hall, Inc. Englewood Cliffs: New Jersey, 1992.
33. Magasin M, Gehlen FL. Unwise Decisions and Unanticipated Consequences. *Sloan Management Review*, Fall 1999.
34. Robert CP. *The Bayesian Choice: A Decision Theoretic Motivation*. Springer-Verlag: New York, 2004.
35. Ghosh M, Carlin BP, Srivastava MS. Probability matching priors for linear calibration. *Test* 1995; **4**:333–357. DOI: 10.1007/BF02562631.
36. Kass RE, Wasserman L. The selection of prior distributions by formal rules. *Journal of the American Statistical Association* 1996; **91**:1343–1370. DOI: 10.1080/01621459.1996.10477003.
37. Lin X, Pittman J, Clarke B. Information Conversion, Effective Samples, and Parameter Size. *IEEE Transactions in Information Theory* 2007; **53**:4438–4456. DOI: 10.1109/TIT.2007.909168.
38. Irony TZ, Singpurwalla ND. Noninformative priors do not exist: a dialogue with José M. Bernardo. *Journal of Statistical Planning and Inference* 1997; **65**:159–189. DOI: 10.1016/S0378-3758(97)00074-8.
39. Tibshirani R. Noninformative priors for one parameter of many. *Biometrika* 1989; **76**:604–608. DOI: 10.1093/biomet/76.3.604.
40. Clarke BS. Implications of reference priors for prior information and for sample size. *Journal of the American Statistical Association* 1996; **91**:173–184. DOI: 10.1080/01621459.1996.
41. Yang R, Berger JO. A catalog of non-informative priors. *Duke University Research Report*, 1998.
42. Gelman A, Rubin DB. Inference from iterative simulation using multiple sequences. *Statistical Science* 1992; **7**:457–472. DOI: 10.1214/ss/1177011136.
43. Brooks SP, Gelman A. General methods for monitoring convergence of iterative simulations. *Journal of Computational and Graphical Statistics* 1998; **7**:434–455. DOI: 10.1080/10618600.1998.10474787.
44. Plummer M, Best N, Cowles K, Vines K. CODA: convergence diagnosis and output analysis for MCMC. *R News* 2006; **6**:7–11.
45. Pasanisi A, Keller M, Parent E. Estimation of a quantity of interest in uncertainty analysis: some help from Bayesian decision theory. *Reliability Engineering and System Safety* 2012; **100**:93–101. DOI: 10.1016/j.res.2012.01.001.
46. Mezzetti M, Ibrahim JG. Bayesian inference for the Cox model using correlated gamma process priors. *Technical report*. Dpt of Biostatistics, Harvard School of Public Health, 2000.
47. James LF. Bayesian calculus for gamma processes with applications to semiparametric intensity models. *Sankhyā* 2003; **65**:179–206. DOI: 10.2307/25053253.
48. Nieto-Barajas LE, Walker SG. Markov beta and gamma processes for modelling hazard rates. *Scandinavian Journal of Statistics* 2002; **29**:413–424. DOI: 10.1111/1467-9469.0029810.1111/1467-9469.00298.
49. Chatzis SP, Korkinof D, Demiris Y. A spatially-constrained normalized gamma process for data clustering. *Artificial Intelligence Applications and Innovations. IFIP Advances in Information and Communication Technology* 2012; **381**:337–346. DOI: 10.1007/978-3-642-33409-2_35.
50. Lindley DV. Reconciliation of probability distributions. *Operations Research* 1983; **31**:866–880. DOI: 10.1287/opre.31.5.866.
51. Press SJ. *Subjective and Objective Bayesian Statistics* (second edition). Wiley: New York, 2003.
52. O'Hagan A, Buck CE, Daneshkhah A, Eiser JR, Garthwaite PH, Jenkinson DJ, Oakley JE, Rakow T. *Uncertain Judgements: Eliciting Expert Probabilities*. John Wiley and Sons: Chichester, 2006.
53. Neal RM. Transferring prior information between models using imaginary data. Technical Report 0108, Dept. Statistics Univ. Toronto, 2001.
54. Morita S, Thall PF, McLuller P. Determining the effective sample size of a parametric prior. *Biometrics* 2008; **64**:595–602. PMC3081791.
55. Artin E. *The Gamma Function*. Holt Rinehart Winston: New York, 1964.
56. Hazewinkel M. Bernoulli Numbers. *Encyclopedia of Mathematics*. Springer, 2001. ISBN 978-1556080104.
57. Park C, Padgett WJ. Accelerated degradation models for failure based on geometric Brownian motion and gamma process. *Lifetime Data Analysis* 2005; **11**:511–527. DOI: 10.1007/s10985-005-5237-8.
58. Bérenguer C, Grall A, Dieulle L, Roussignol M. Maintenance policy for a continuously monitored deteriorating system. *Probability in the Engineering and Informational Science* 2003; **17**:235–250. DOI: 10.1017/S0269964803172063.
59. Mercier S, Bordes L, Remy E, Dautrême E. A stochastic model for competing degradations. *Proceedings of the 22nd ESREL Conference*, Amsterdam, 2013.
60. Stehlik M. Exact likelihood ratio scale and homogeneity testing of some loss processes. *Statistics and Probability Letters* 2006; **76**:19–26. DOI: 10.1016/j.spl.2005.06.005.
61. Grall-Maës E. Use of the Kolmogorov-Smirnov test for gamma process. *Proceedings of the Institution of Mechanical Engineers, Part O: Journal of Risk and Reliability*, vol. 6, 2012, 624–634.
62. Berger JO. *Statistical Decision Theory and Bayesian Analysis*. Springer-Verlag: New York, 1985.
63. Tversky A, Kahneman D. Judgement under uncertainty: Heuristics and biases. *Science* 1974; **185**:1124–1131. DOI: 0.1126/science.185.4157.1124.
64. Celeux G, Marin JM, Robert CP. Sélection bayésienne de variables en régression linéaire. *Journal de la Société Française de Statistiques* 2006; **147**:59–80. (in French).

65. De Santis F, Mortera J, Nardi A. Jeffreys priors for survival models with censored data. *Journal of Statistical Planning and Inference* 2001; **99**:193–209. DOI: 10.1016/S0378-3758(01)00080-5.
66. Lindley DV. Bayesian Statistics: A Review. *SIAM*: Philadelphia, PA, 1972.
67. Dongliang L. Estimation of stochastic degradation models using uncertain inspection data. *Ph.D. thesis*, University of Waterloo, Canada, 2012.
68. Keller M, Bousquet N. Estimation of a flaw size distribution from a data mixture based on destructive tests and non-destructive in-service inspections. *Proceedings of the ESREL-PSAM Conference*, Chapman & Hall/CRC, Taylor & Francis Group, Helsinki, 2012.
69. Consonni G, Gutiérrez-Peña E, Veronese P. Compatible priors for Bayesian model comparison with an application to the Hardy-Weinberg equilibrium model. *Test* 2008; **17**:585–605. DOI: 10.1007/s11749-007-0057-7.